

LCC: a lineshape-correlation coefficient for 2D HSQC spectral similarity

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Abstract

Motivation: Quantitative comparison of two-dimensional HSQC spectra underlies ligand screening and molecular fingerprinting, yet the established similarity measures each fail in one of two opposite ways. On a dense protein ^1H - ^{15}N amide fingerprint, bin-histogram methods lose margin under the small peak drifts a titration induces, while tree and nearest-neighbour methods saturate because in a crowded fingerprint every peak finds a coincidental near neighbour. A measure is needed that tolerates small physical drifts yet still penalizes genuinely relocated intensity.

Results: We introduce the Lineshape Correlation Coefficient (LCC), which renders each spectrum onto a shared grid, blurs it by the physical linewidth plus the expected shift drift, and scores the two images by their mean-centred normalized cross-correlation at zero lag. The Gaussian render gives graded shift tolerance while mean-centring penalizes intensity that is not co-located. On two independent benchmarks — a dense protein ^1H - ^{15}N titration (same protein vs a different protein) and a sparse small-molecule ^1H - ^{13}C set (six compounds each recorded twice, 6 same-compound vs 60 different-compound pairs) — LCC gives the widest same/different separation of all methods tested (0.75 and 0.81); the tree and nearest-neighbour methods leave the two classes overlapping (margin ≤ 0.04) in both regimes, and the bin method trails (separation 0.29 and 0.69). Removing the mean-centring drops the protein separation from 0.75 to 0.59, confirming it as the discriminating step.

Availability and implementation: Freely available under the MIT license at https://github.com/deepnmr/spectra_similarity implemented in Python 3 (NumPy; Matplotlib optional for figures); platform independent. The repository includes the three reference methods, both benchmark harnesses and a test suite.

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1 Introduction

A recurring task in NMR is to score how similar two 2D HSQC spectra are: tracking a ^1H - ^{15}N amide fingerprint across a ligand titration, mapping a chemical shift perturbation on binding (Shuker *et al.*, 1996; Pellecchia *et al.*, 2008; Erlanson *et al.*, 2016; Williamson, 2013), or matching a small-molecule ^1H - ^{13}C spectrum against a database — a comparison central to 2D-NMR metabolomics and natural-product work (Robinette *et al.*, 2012; Bingol and Brüscheweiler, 2014; Worley and Powers, 2013), where spectra are also matched against community libraries (Ulrich *et al.*, 2008; Steinbeck and Kuhn, 2004; Wishart *et al.*, 2022). Three families of measure are in use. The bin method of Bodis *et al.* integrates each spectrum over an $n \times n$ grid at a range of resolutions and reports a weighted Jaccard (Ružička) index (Bodis *et al.*, 2007, 2009), in the lineage of the binning and normalization schemes used across NMR metabonomics (Craig *et al.*, 2006; Dieterle *et al.*, 2006). The tree method of Castillo *et al.* encodes a spectrum as a recursive centre-of-mass quad-tree and compares the trees node-by-node with a shift-tolerant node score (Castillo *et al.*, 2013). The

peak method of Pierens *et al.* picks peaks and matches each to its nearest neighbour in the other spectrum (Pierens *et al.*, 2012).

These fail in two opposite ways. Bin histograms are **shift-brittle**: a peak straddling a bin edge splits its intensity across neighbouring bins, so a small drift erodes the score. The tree and nearest-neighbour measures are **shift-blind**: when many peaks share a region every peak has some near neighbour and every spectrum a similar mass-centre structure, so both saturate and retain little contrast. A measure for real HSQC comparison must tolerate small physical drifts yet still penalize genuinely relocated intensity.

2 Implementation

LCC keeps the physically meaningful ingredients of the three methods — area-weighted integration and unit normalization (Bodis), a physical shift tolerance (Castillo), a lineshape picture of the spectrum (Pierens) — and adds a mean-centred correlation as the discriminating step. Given two spectra (processed Bruker data from any standard pipeline — Delaglio *et al.*, 1995; Vranken *et al.*, 2005; Skinner *et al.*, 2016; Ludwig and Günther, 2011 — or peak lists) compared inside a common window, it proceeds in three steps.

Render. Negative intensities are clipped, each point is weighted by its local integration area, and both spectra are histogrammed onto a **single shared grid** so they are directly comparable irrespective of their native digital resolution.

Blur. Each image is convolved with a separable Gaussian whose width per axis is the physical linewidth combined with the expected drift, $\sigma = \sqrt{\ell^2 + d^2}$. This replaces the bin method’s hard edges with a continuous response: for an isolated peak on a large grid (where mean-centring is negligible) two copies displaced by (Δ_1, Δ_2) correlate as $\exp(-\Delta_1^2/4\sigma_1^2 - \Delta_2^2/4\sigma_2^2)$, smooth and monotone in the drift, so a physical shift costs little and a random relocation costs a lot.

Score. Each image is mean-centred and the similarity is the zero-lag normalized cross-correlation (the Pearson coefficient) of the two centred images, clamped to $[0, 1]$ — the spectral-imaging analogue of the cosine/contrast-angle similarity long used in mass-spectral library search (Stein and Scott, 1994; Wan *et al.*, 2002). Mean-centring is the discriminating step: a cell where one spectrum has a peak and the other is empty contributes a negative product and lowers the score, so intensity that is not co-located is actively penalized rather than, as in the nearest-neighbour distance, ignored. No shift search is performed — aligning the images (as in NMR peak-alignment tools such as icoshift; Savorani *et al.*, 2010) would let a different spectrum slide into registration and re-saturate, the failure mode of the tree and nearest-neighbour methods.

Self-similarity is exactly one (as it is for all methods tested); the score is symmetric exactly, and monotone in the drift of an isolated peak (verified in the unit tests). The implementation is pure NumPy (Harris *et al.*, 2020), reads processed Bruker data directly (cf. Helmus and Jaroniec, 2013), needs no peak picking and no external solver, and runs in about 6 ms per spectrum pair on a 250×350 grid (Apple Silicon, single core); a 19-test suite (all passing) checks self-similarity, related-beats-unrelated ordering and monotonicity in drift. Figures use Matplotlib (Hunter, 2007). A full derivation with proofs is provided as Supplementary Material.

3 Results

We benchmarked LCC against the three reference methods on **two independent, opposite regimes**. Discrimination is summarized by the **separation** (mean same-class score — mean different-class score) and the **margin** (worst same-class — best different-class score, what a classification threshold sees); both are on each method’s own $[0, 1]$ scale (Table 1; Fig. 1). All methods self-score exactly 1.

Dense protein ^1H - ^{15}N . The reference is the apo spectrum of the phosphatase PRL3, compared with six spectra of the same protein along a ligand titration (should score high) and with a **different** protein (should score low), inside ^1H 6.5–10 $\times$ ^{15}N 105–130 ppm. LCC scores the same protein 0.94 (range 0.89–0.97) and the different protein 0.18, a margin of 0.71 — three times the bin method’s 0.24, the strongest baseline. Mean-centring earns that gap: an un-centred cosine correlation on the same images drops the separation from 0.75 to 0.59. The gain is not a tuned parameter (it holds across the physical blur range; Supplementary).

Sparse small-molecule ^1H - ^{13}C . To test the opposite regime — and to replace the single negative spectrum above with a large negative set — we used public HSQC peak lists (simpleNMR; Data availability) for six compounds each recorded twice, and asked each method to score the 6 same-compound pairs above the 60 different-compound pairs, inside ^1H 0–10 $\times$ ^{13}C 0–165 ppm. LCC again leads (mean same 0.82, mean different 0.003, separation 0.81). The bin method is far stronger here (0.69) than on the dense fingerprint (0.29) — with few, well-separated peaks its hard bins rarely straddle a peak — while the tree and nearest-neighbour methods **leave the two classes barely separable in this regime too** (margins 0.00 and 0.04, despite separations of 0.39 and 0.10): over a wide window a different molecule still has a near neighbour for every peak. Their shift tolerance is genuine (per-pair, it keeps two solvent-shifted same-compound recordings high, 0.83–1.00, where the position-sensitive LCC and bin methods drop to 0.30–0.60) but it tolerates *everything*, so it does not become discrimination; LCC recovers that tolerance controllably through its blur width.

Table 1: Separation and worst-case margin for all methods on both regimes, computed from unrounded scores. Every method self-scores 1.00.

Method	^1H - ^{15}N sep	margin	^1H - ^{13}C sep	margin
Bin (Bodis, 2009)	0.29	0.24	0.69	0.20
Bin + 45° rotation	0.25	0.20	0.69	0.18
Quad-tree (Castillo, 2013)	0.03	−0.01	0.39	0.00
Nearest neighbour (Pierens, 2012)	0.04	0.03	0.10	0.04
LCC (this work)	0.75	0.71	0.81	0.37

LCC leads in both regimes — the only method that keeps same-class similarity high while pushing different-class similarity down, whether the fingerprint is a dense protein amide region or a handful of scattered small-molecule crosspeaks. Its two knobs are the per-axis blur widths, both physical: the linewidth sets the floor and the expected shift perturbation sets how much drift is tolerated. LCC measures position-and-intensity coincidence, not molecular identity, so a close analogue will score intermediate — correct for a similarity, a caveat for hard classification. The three reference methods are our own reimplementations, and the ^1H - ^{13}C spectra are rendered from peak lists inside one common window, so the *ranking*, not the exact values, is the result.

Data availability

The dense ^1H - ^{15}N benchmark uses PRL3 and OAA Bruker spectra available from the author on reasonable request. The sparse ^1H - ^{13}C benchmark uses public HSQC peak lists from the simpleNMR example set (Hughes and Kenwright, 2024; <https://github.com/EricHughesABC/simpleNMR>), downloaded automatically by `bench_13c.py`. Both harnesses (`bench.py`, `bench_13c.py`) and the tabulated scores are in the repository.

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Conflict of interest

None declared.

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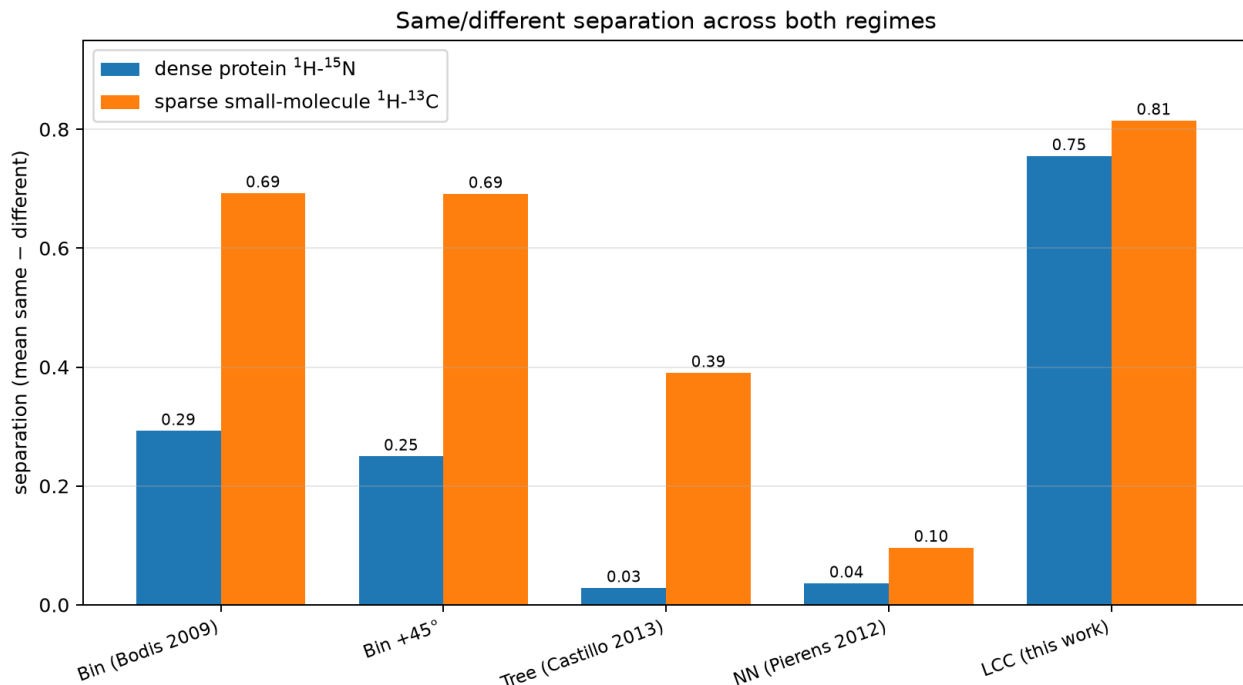


Figure 1: **Figure 1.** Same/different separation (mean same-class — mean different-class similarity) for all five methods on the two benchmarks: dense protein $^1\text{H}-^{15}\text{N}$ (blue) and sparse small-molecule $^1\text{H}-^{13}\text{C}$ (orange). LCC is the only method with a large separation in both regimes; the tree and nearest-neighbour methods saturate in both.

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